

Causal and Bayesian Networks

In this chapter we introduce causal networks, which are the basic graphical feature for (almost) everything in this book. We give rules for reasoning about relevance in causal networks; is knowledge of A relevant for my belief about B ? These sections deal with reasoning under uncertainty in general. Next, Bayesian networks are defined as causal networks with the strength of the causal links represented as conditional probabilities. Finally, the chain rule for Bayesian networks is presented. The chain rule is the property that makes Bayesian networks a very powerful tool for representing domains with inherent uncertainty. The sections on Bayesian networks assume knowledge of probability calculus as laid out in Sections 1.1–1.4.

2.1 Reasoning Under Uncertainty

2.1.1 Car Start Problem

The following is an example of the type of reasoning that humans do daily.

“In the morning, my car will not start. I can hear the starter turn, but nothing happens. There may be several reasons for my problem. I can hear the starter roll, so there must be power from the battery. Therefore, the most-probable causes are that the fuel has been stolen overnight or that the spark plugs are dirty. It may also be due to dirt in the carburetor, a loose connection in the ignition system, or something more serious. To find out, I first look at the fuel meter. It shows half full, so I decide to clean the spark plugs.”

To have a computer do the same kind of reasoning, we need answers to questions such as, “What made me conclude that among the probable causes “stolen fuel”, and “dirty spark plugs” are the two most-probable causes?” or “What made me decide to look at the fuel meter, and how can an observation concerning fuel make me conclude on the seemingly unrelated spark plugs?” To be more precise, we need ways of representing the problem and ways of

performing inference in this representation such that a computer can simulate this kind of reasoning and perhaps do it better and faster than humans.

For propositional logic, Boolean logic is the representation framework, and various derived structures, such as truth tables and binary decision diagrams, have been invented together with efficient algorithms for inference.

In logical reasoning, we use four kinds of logical connectives: conjunction, disjunction, implication, and negation. In other words, simple logical statements are of the kind, “if it rains, then the lawn is wet,” “both John and Mary have caught the flu,” “either they stay at home or they go to the cinema,” or “the lawn is not wet.” From a set of logical statements, we can deduce new statements. From the two statements “if it rains, then the lawn is wet” and “the lawn is not wet,” we can infer that it is not raining.

When we are dealing with uncertain events, it would be nice if we could use similar connectives with certainties rather than truth values attached, so we may extend the truth values of propositional logic to “certainties,” which are numbers between 0 and 1. A certainty 0 means “certainly not true,” and the higher the number, the higher the certainty. Certainty 1 means “certainly true.”

We could then work with statements such as, “if I take a cup of coffee while on break, I will with certainty 0.5 stay awake during the next lecture” or “if I take a short walk during the break, I will with certainty 0.8 stay awake during the next lecture.” Now, suppose I take a walk as well as have a cup of coffee. How certain can I be to stay awake? To answer this, I need a rule for how to *combine* certainties. In other words, I need a function that takes the two certainties 0.5 and 0.8 and returns a number, which should be the certainty resulting from combining the certainty from the two statements.

The same is needed for chaining: “if a then b with certainty x ,” and “if b then c with certainty y .” I know a , so what is the certainty of c ?

It has turned out that any function for combination and chaining will in some situations lead to wrong conclusions.

Another problem, which is also a problem for logical reasoning, is abduction: I have the rule “a woman has long hair with certainty 0.7.” I see a long-haired person. What can I infer about the person’s sex?

2.1.2 A Causal Perspective on the Car Start Problem

A way of structuring a situation for reasoning under uncertainty is to construct a graph representing causal relations between events.

Example 2.1 (A reduced Car Start Problem).

To simplify the situation, assume that we have the events $\{yes, no\}$ for *Fuel?*, $\{yes, no\}$ for *Clean Spark Plugs?*, $\{full, \frac{1}{2}, empty\}$ for *Fuel Meter*, and $\{yes, no\}$ for *Start?*. In other words, the events are clustered around variables, each with a set of outcomes, also called *states*. We know that the state of *Fuel?* and the state of *Clean Spark Plugs?* have a causal impact on

the state of *Start?*. Also, the state of *Fuel?* has an impact on the state of *Fuel Meter Standing*. This is represented by the graph in Figure 2.1.

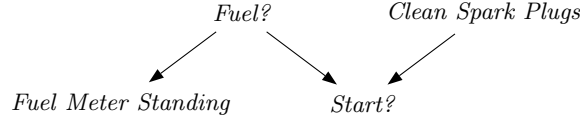


Fig. 2.1. A causal network for the reduced Car Start Problem.

If we add a direction from *no* to *yes* inside each variable (and from *empty* to *full*), we can also represent directions of the impact. For the present situation, we can say that all the impacts are positive (with the direction); that is, the more the certainty of the cause is moved in a positive direction, the more the certainty of the affected variable will also be moved in a positive direction. To indicate this, we can label the links with the sign “+” as is done in Figure 2.2.

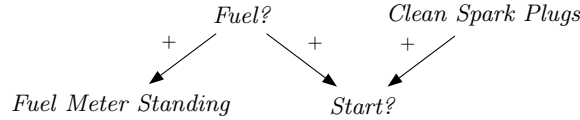


Fig. 2.2. A causal network for the reduced Car Start Problem with a sign indicating direction of impact.

We can use the graph in Figure 2.2 to perform some reasoning. Obviously, if I know that the spark plugs are not clean, then the certainty for no start will increase. However, my situation is the opposite. I realize that I have a start problem. As my certainty on *Start?* is moved in a negative direction, I find the possible causes (*Clean Spark Plugs?* and *Fuel?*) for such a move more certain; that is, the sign “+” is valid for both directions. Now, because the certainty on for *Fuel?* = *no* has increased, I will have a higher expectation that *Fuel Meter Standing* is in state *empty*.

The movement of the certainty for *Fuel Meter Standing* tells me that by reading the fuel meter I will get information related to the start problem. I read the fuel meter, it says $\frac{1}{2}$, and reasoning backward yields that the certainty on *Fuel?* is moved in a negative direction.

So far, the reasoning has been governed by simple rules that can easily be formalized. The conclusion is harder: “Lack of fuel does not seem to be the reason for my start problem, so most probably the spark plugs are not clean.” Is there a formalized rule that allows this kind of reasoning on a causal

network to be computerized? We will return to this problem in Section 2.2.

Note: The reasoning has focused on changes of certainty. In certainty calculus, if the actual certainty of a specific event must be calculated, then knowledge of certainties prior to any information is also needed. In particular, prior certainties are required for the events that are not effects of causes in the network. If, for example, my car cannot start, the actual certainty that the fuel has been stolen depends on my neighborhood.

2.2 Causal Networks and d-Separation

A causal network consists of a set of *variables* and a set of *directed links* (also called *arcs*) between variables. Mathematically, the structure is called a *directed graph*. When talking about the relations in a directed graph, we use the wording of family relations: if there is a link from A to B , we say that B is a *child* of A , and A is a *parent* of B .

The variables represent propositions (or sample spaces), see also Section 1.3. A variable can have any number of states (or outcomes). A variable may, for example, be the color of a car (states *blue*, *green*, *red*, *brown*), the number of children in a specific family (states 0, 1, 2, 3, 4, 5, 6, > 6), or a disease (states *bronchitis*, *tuberculosis*, *lung cancer*). Variables may have a countable or a continuous state set, but we consider only variables with a finite number of states (we shall return to the issue of continuous state spaces in Section 3.3.8).

In a causal network, a variable represents a set of possible states of affairs. A variable is in exactly one of its states; which one may be unknown to us.

As illustrated in Section 2.1.2, causal networks can be used to follow how a change of certainty in one variable may change the certainty for other variables. We present in this section a set of rules for that kind of reasoning. The rules are independent of the particular calculus for uncertainty.

Serial Connections

Consider the situation in Figure 2.3. Here A has an influence on B , which in turn has an influence on C . Obviously, evidence about A will influence the certainty of B , which then influences the certainty of C . Similarly, evidence about C will influence the certainty of A through B . On the other hand, if the state of B is known, then the channel is blocked, and A and C become independent; we say that A and C are d-separated given B . When the state of a variable is known, we say that the variable is *instantiated*.

We conclude that evidence may be transmitted through a serial connection unless the state of the variable in the connection is known.

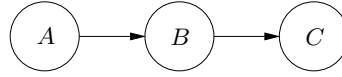


Fig. 2.3. Serial connection. When B is instantiated, it blocks communication between A and C .

Example 2.2. Figure 2.4 shows a causal model for the relations between *Rainfall* (*no, light, medium, heavy*), *Water level* (*low, medium, high*), and *Flooding* (*yes, no*). If I have not observed the water level, then knowing that there has been a flooding will increase my belief that the water level is high, which in turn will tell me something about the rainfall. The same line of reasoning holds in the other direction. On the other hand, if I already know the water level, then knowing that there has been flooding will not tell me anything new about rainfall.



Fig. 2.4. A causal model for *Rainfall*, *Water level*, and *Flooding*.

Diverging Connections

The situation in Figure 2.5 is called a *diverging* connection. Influence can pass between all the children of A unless the state of A is known. That is, $B, C, \dots, E$ are d-separated given A .

Evidence may be transmitted through a diverging connection unless it is instantiated.

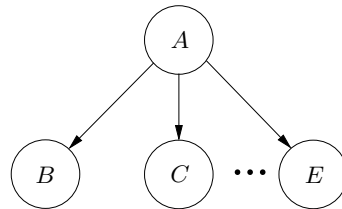


Fig. 2.5. Diverging connection. If A is instantiated, it blocks communication between its children.

Example 2.3. Figure 2.6 shows the causal relations between *Sex* (*male, female*), *length of hair* (*long, short*), and *stature* (<168 cm, ≥ 168 cm).

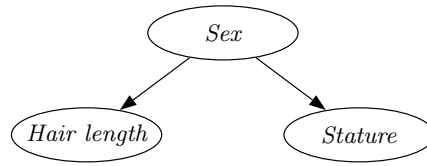


Fig. 2.6. Sex has an impact on length of hair as well as stature.

If we do not know the sex of a person, seeing the length of his/her hair will tell us more about the sex, and this in turn will focus our belief on his/her stature. On the other hand, if we know that the person is a man, then the length of his hair gives us no extra clue on his stature.

Converging Connections

A description of the situation in Figure 2.7 requires a little more care. If nothing is known about A except what may be inferred from knowledge of its parents $B, \dots, E$, then the parents are independent: evidence about one of them cannot influence the certainties of the others through A . Knowledge of one possible cause of an event does not tell us anything about the other possible causes. However, if anything is known about the consequences, then information on one possible cause may tell us something about the other causes. This is the *explaining away* effect illustrated in the car start problem: the car cannot start, and the potential causes include dirty spark plugs and an empty fuel tank. If we now get the information that there is fuel in the tank, then our certainty in the spark plugs being dirty will increase (since this will explain why the car cannot start). Conversely, if we get the information that there is no fuel on the car, then our certainty in the spark plugs being dirty will decrease (since the lack of fuel explains why the car cannot start). In Figure 2.8, two examples are shown. Observe that in the second example we observe only A indirectly through information about F ; knowing the state of F tells us something about the state of E , which in turn tells us something about A .

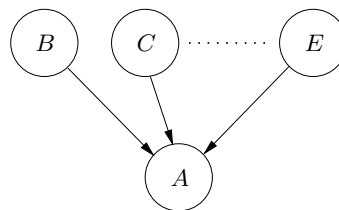


Fig. 2.7. Converging connection. If A changes certainty, it opens communication between its parents.

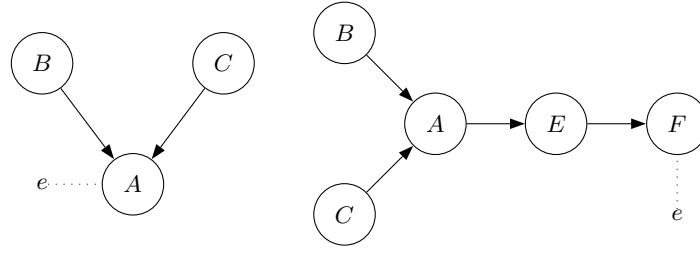


Fig. 2.8. Examples in which the parents of A are dependent. The dotted lines indicate insertion of evidence.

The conclusion is that evidence may be transmitted through a converging connection only if either the variable in the connection or one of its descendants has received evidence.

Remark: Evidence about a variable is a statement of the certainties of its states. If the variable is instantiated, we call it *hard* evidence; otherwise, it is called *soft*. In the example above, we can say that hard evidence about the variable F provides soft evidence about the variable A . Blocking in the case of serial and diverging connections requires hard evidence, whereas opening in the case of converging connections holds for all kinds of evidence.

Example 2.4. Figure 2.9 shows the causal relations among *Salmonella* infection, *flu*, *nausea*, and *pallor*.

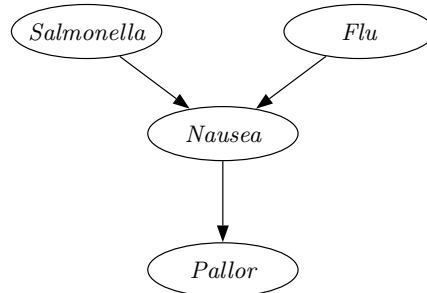


Fig. 2.9. Salmonella and flu may cause nausea, which in turn causes pallor.

If we know nothing of nausea or pallor, then the information on whether the person has a *Salmonella* infection will not tell us anything about flu. However, if we have noticed that the person is pale, then the information that he/she does not have a *Salmonella* infection will make us more ready to believe that he/she has the flu.

2.2.1 d-separation

The three preceding cases cover all ways in which evidence may be transmitted through a variable, and following the rules it is possible to decide for any pair of variables in a causal network whether they are independent given the evidence entered into the network. The rules are formulated in the following definition.

Definition 2.1 (d-separation). *Two distinct variables A and B in a causal network are d-separated (“d” for “directed graph”) if for all paths between A and B , there is an intermediate variable V (distinct from A and B) such that either*

- *the connection is serial or diverging and V is instantiated*
- or*
- *the connection is converging, and neither V nor any of V ’s descendants have received evidence.*

If A and B are not d-separated, we call them d-connected.

Figure 2.10 gives an example of a larger network. The evidence entered at B and M represents instantiations. If evidence is entered at A , it may be transmitted to D . The variable B is blocked, so the evidence cannot pass through B to E . However, it may be passed to H and K . Since the child M of K has received evidence, evidence from H may pass to I and further to E, C, F, J , and L , so the path $A - D - H - K - I - E - C - F - J - L$ is a d-connecting path. Figure 2.11 gives two other examples.

Note that although A and B are d-connected, changes in the belief in A will not necessarily change the belief in B . To stress this difference, we will sometimes say that A and B are *structurally independent* if they are d-separated (see also Exercise 2.23).

In connection to d-separation, a special set of nodes for a node A is the so-called *Markov blanket* for A :

Definition 2.2. *The Markov blanket of a variable A is the set consisting of the parents of A , the children of A , and the variables sharing a child with A .*

The Markov blanket has the property that when instantiated, A is d-separated from the rest of the network (see Figure 2.12).

You may wonder why we have introduced d-separation as a definition rather than as a theorem. A theorem should be as follows.

Claim: If A and B are d-separated, then changes in the certainty of A have no impact on the certainty of B .

However, the claim cannot be established as a theorem without a more-precise description of the concept of “certainty.” You can take d-separation as a property of human reasoning and require that any certainty calculus should comply with the claim.

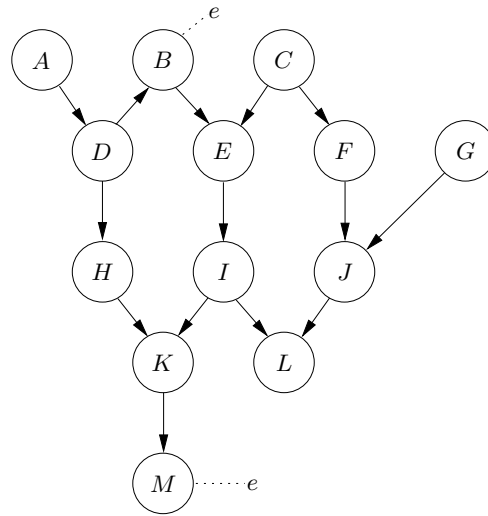


Fig. 2.10. A causal network with M and B instantiated. The node A is d-separated from G only.

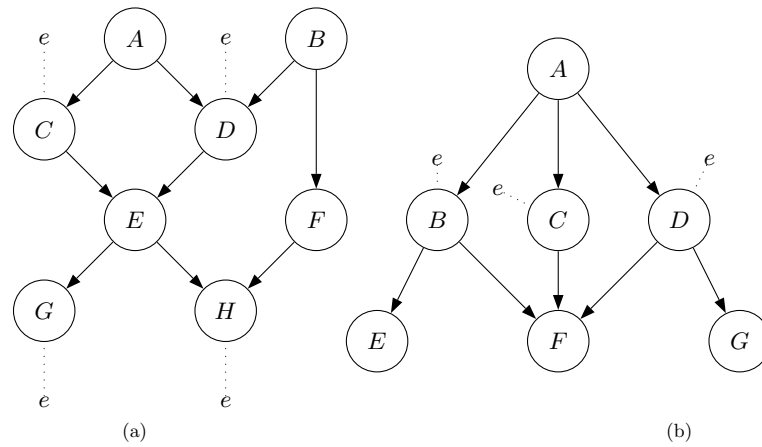


Fig. 2.11. Causal networks with hard evidence entered (the variables are instantiated). (a) Although all neighbors of E are instantiated, it is d-connected to F , B , and A . (b) F is d-separated from the remaining uninstantiated variables.

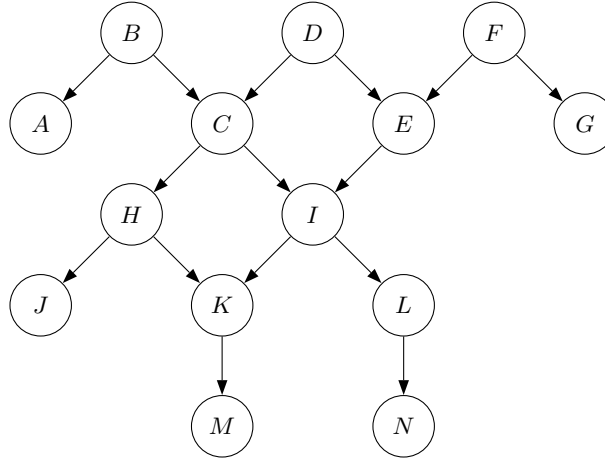


Fig. 2.12. The Markov blanket for I is $\{C, E, H, K, L\}$. Note that if only I 's neighbors are instantiated, then J is not d -separated from I .

From the definition of d -separation we see that in order to test whether two variables, say A and B , are d -separated given hard evidence on a set of variables $\mathcal{C}$ you would have to check whether all paths connecting A and B are d -separating paths. An easier way of performing this test, without having to consider the various types of connections, is as follows: First you construct the so-called *ancestral graph* consisting of A , B , and $\mathcal{C}$ together with all nodes from which there is a directed path to either A , B , or $\mathcal{C}$ (see Figure 2.13(a)). Next, you insert an undirected link between each pair of nodes with a common child and then you make all links undirected. The resulting graph (see Figure 2.13(b)) is known as the *moral graph* for Figure 2.13(a). The moral graph can now be used to check whether A and B are d -separated given $\mathcal{C}$: if all paths connecting A and B intersect $\mathcal{C}$, then A and B are d -separated given $\mathcal{C}$.

The above procedure generalizes straightforwardly to the case in which we work with sets of variables rather than single variables: you just construct the ancestral graph using these sets of variables and perform the same steps as above: $\mathcal{A}$ and $\mathcal{B}$ are then d -separated given $\mathcal{C}$ if all paths connecting a variable in $\mathcal{A}$ with a variable in $\mathcal{B}$ intersect a variable in $\mathcal{C}$.

2.3 Bayesian Networks

2.3.1 Definition of Bayesian Networks

Causal relations also have a quantitative side, namely their *strength*. This can be expressed by attaching numbers to the links.